Balances transport cost ($\|v_t\|^2$) with a diffusion penalty ($\|\nabla \log \rho_t\|^2$)

speed^2 of particle at location x ~ kinetic energy

Diffusion 'penalty' ~ high if ρ_t peaked around x, low if flat

Schrödinger Problem

- Another way to arrive at this entropic interpolation ρ_t^ε is by directly regularizing Benamou-Brenier:

speed^2 of particle at location x ~ kinetic energy

$$\min_{(\alpha_t, v_t)_t \text{ s.t. [Cont.Eq.]}} \int_0^1 \int_{\mathbb{R}^d} \left(\|v_t(x)\|^2 + \frac{\varepsilon}{4} \|\nabla \log(\rho_t)(x)\|^2 \right) \mathrm{d}\rho_t(x)\,\mathrm{d}t$$

Diffusion 'penalty' ~ high if ρ_t peaked around x, low if flat

- Intuition:
  - **Balances transport cost $(\|v_t\|^2)$ with a diffusion penalty ($\|\nabla \log \rho_t\|^2$).**
  - Mass moves along velocity fields but is also allowed to spread like a heat flow.
  - Interpolating between sharp OT geodesics and noisy Schrödinger bridges.

ML Applications